

Yes 👍

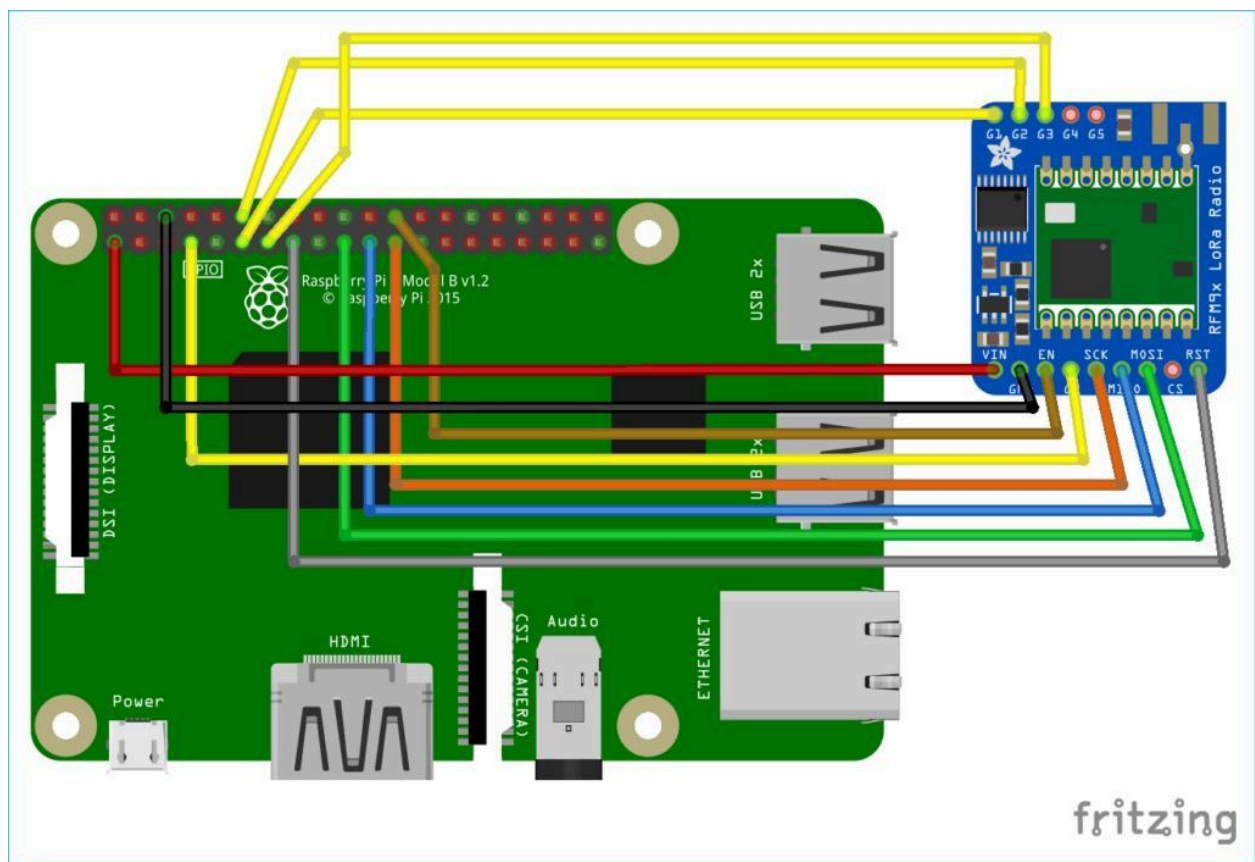
You can easily send messages up to **2 km** using:

- Raspberry Pi 5
- SX126X LoRa HAT

LoRa is made for long-distance communication.

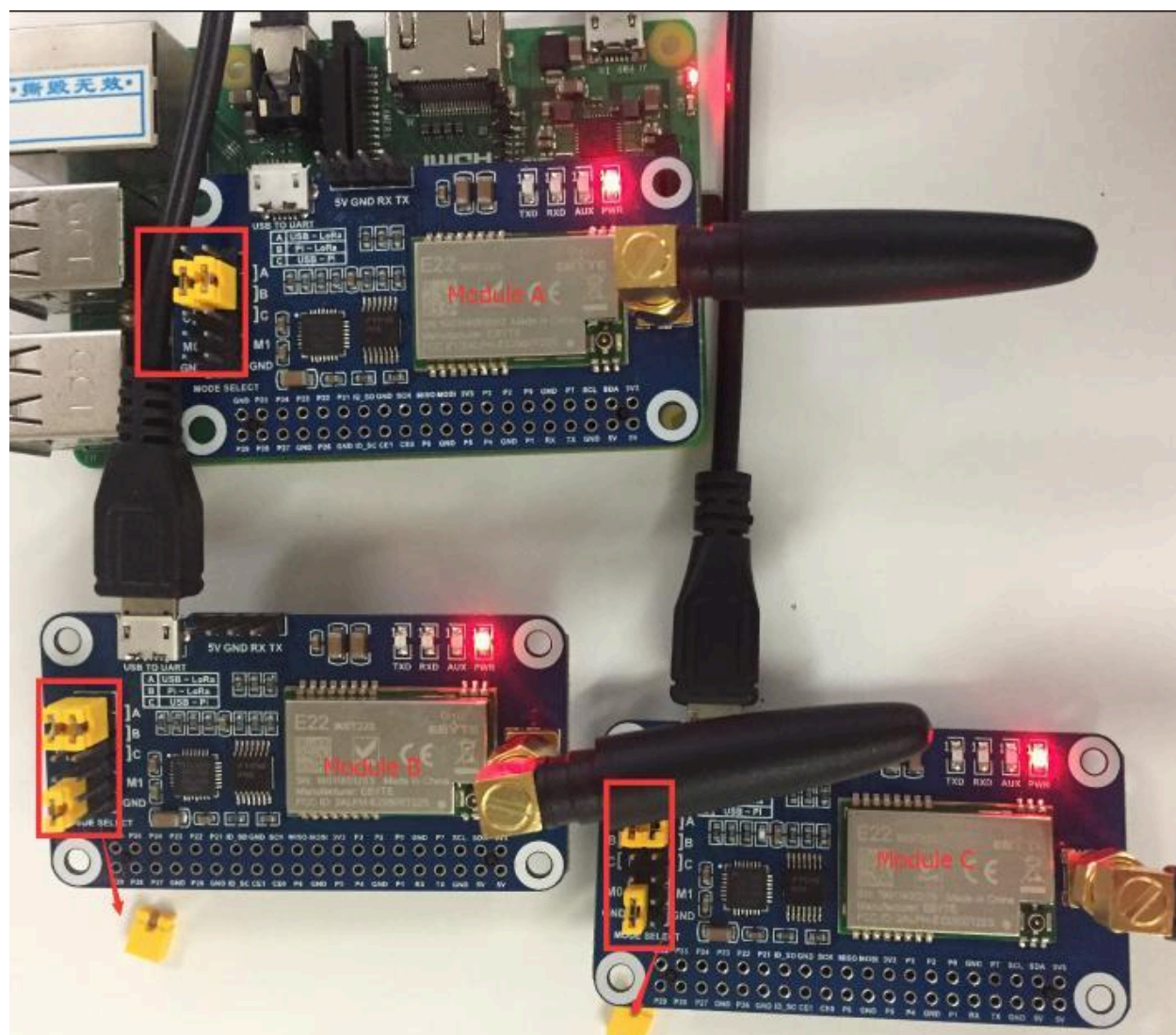
In open area, 2 km is very easy. ([Zbotic](#))

What You Need





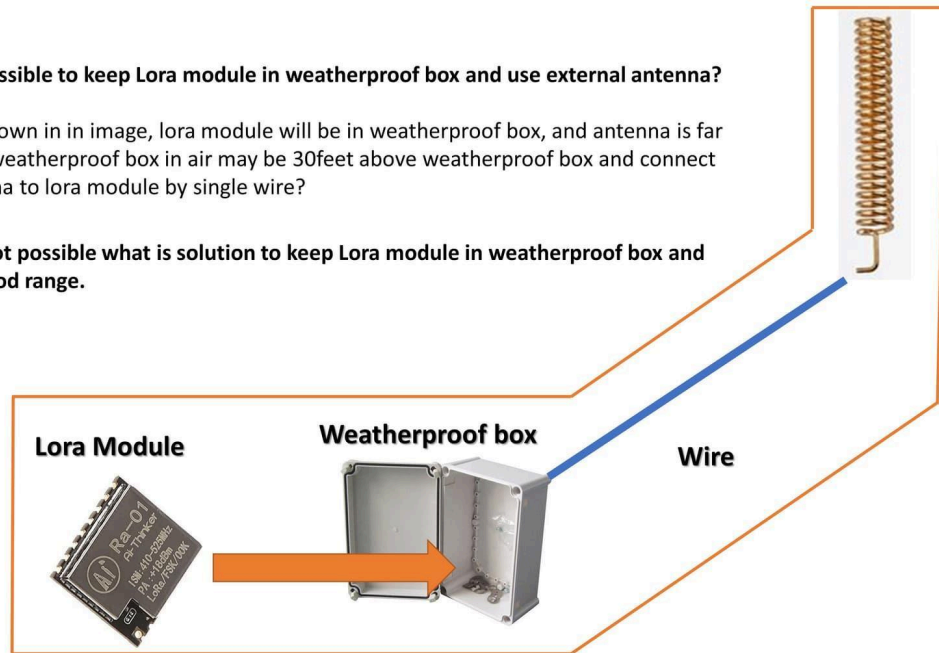




Is it possible to keep Lora module in weatherproof box and use external antenna?

Like shown in in image, lora module will be in weatherproof box, and antenna is far From weatherproof box in air may be 30feet above weatherproof box and connect Antenna to lora module by single wire?

If its not possible what is solution to keep Lora module in weatherproof box and Get good range.





- ## Hardware

STEP 1 — Enable SPI

In Raspberry Pi terminal:

```
sudo raspi-config
```

Go to:

Interface Options

→ SPI

→ Enable

Then reboot:

```
sudo reboot
```

STEP 2 — Install Python Library

```
sudo apt update  
pip install sx126x
```

([PyPI](#))

STEP 3 — Sender Code

Create file:

```
nano sender.py
```

Paste:

```
from sx126x import SX126x  
import time
```



```
# Initialize LoRa
lora = SX126x()

print("Sender Started")

while True:
    msg = "Hello from Raspberry Pi 5"

    lora.send(msg.encode())

    print("Sent:", msg)

    time.sleep(5)
```

Save:

CTRL + X
Y
ENTER

Run:

```
python3 sender.py
```

STEP 4 — Receiver Code

Create:

```
nano receiver.py
```

Paste:

```
from sx126x import SX126x

lora = SX126x()

print("Receiver Started")

while True:
```

```
data = lora.receive()

if data:
    print("Received:", data.decode())
```

Run:

```
python3 receiver.py
```

STEP 5 — Important Settings

Both LoRa HATs MUST use:

- Same frequency
- Same spreading factor
- Same bandwidth

For India use:

865 MHz

([Zbotic](#))

HOW TO GET 2 KM RANGE

Important Tips

1. Use Good Antenna

Biggest factor for range.

2. Keep Antenna High

3–5 meter height improves range a lot. ([Zbotic](#))

3. Open Area

No buildings/trees.

4. Use Lower Data Rate

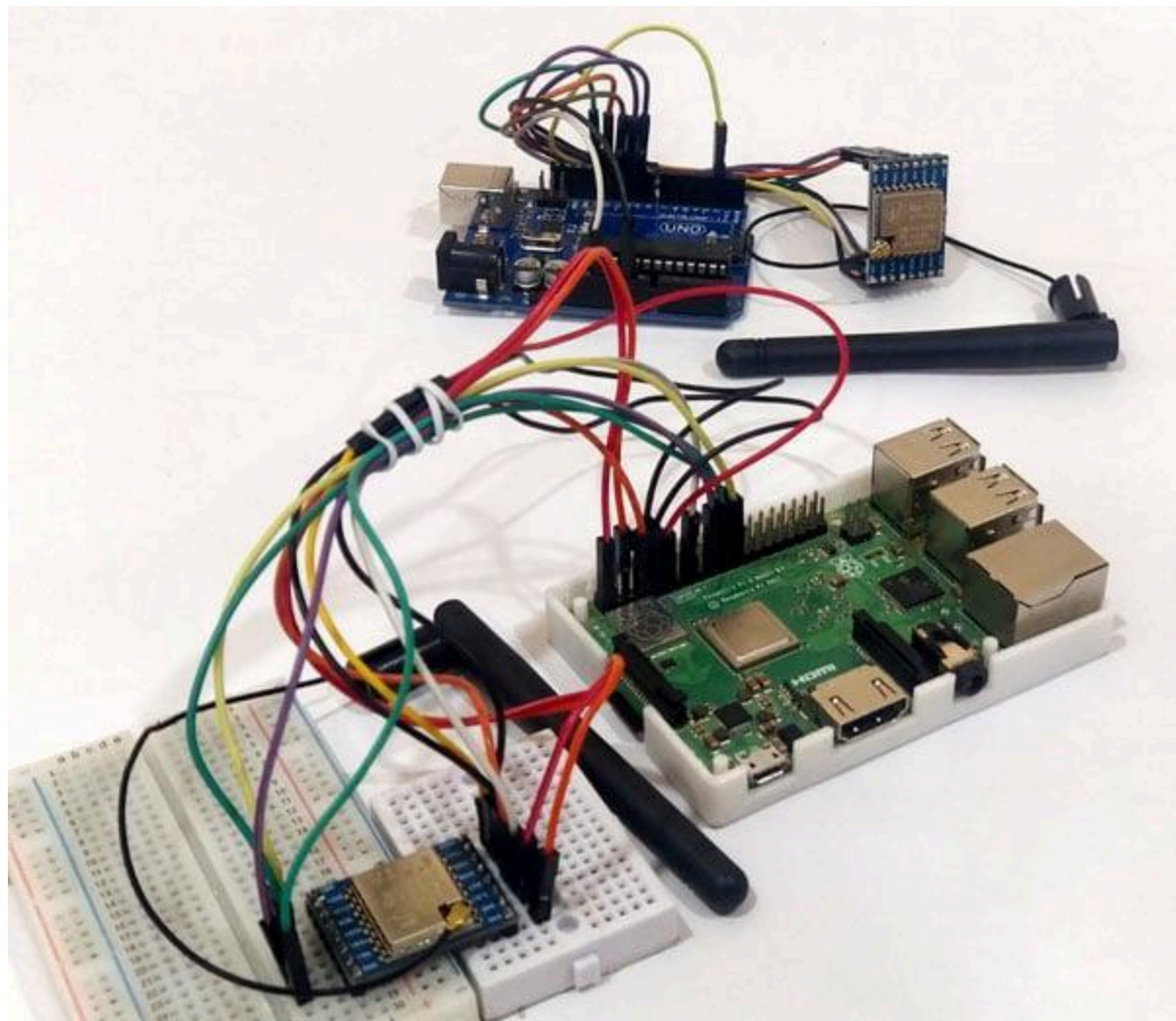
Longer range.

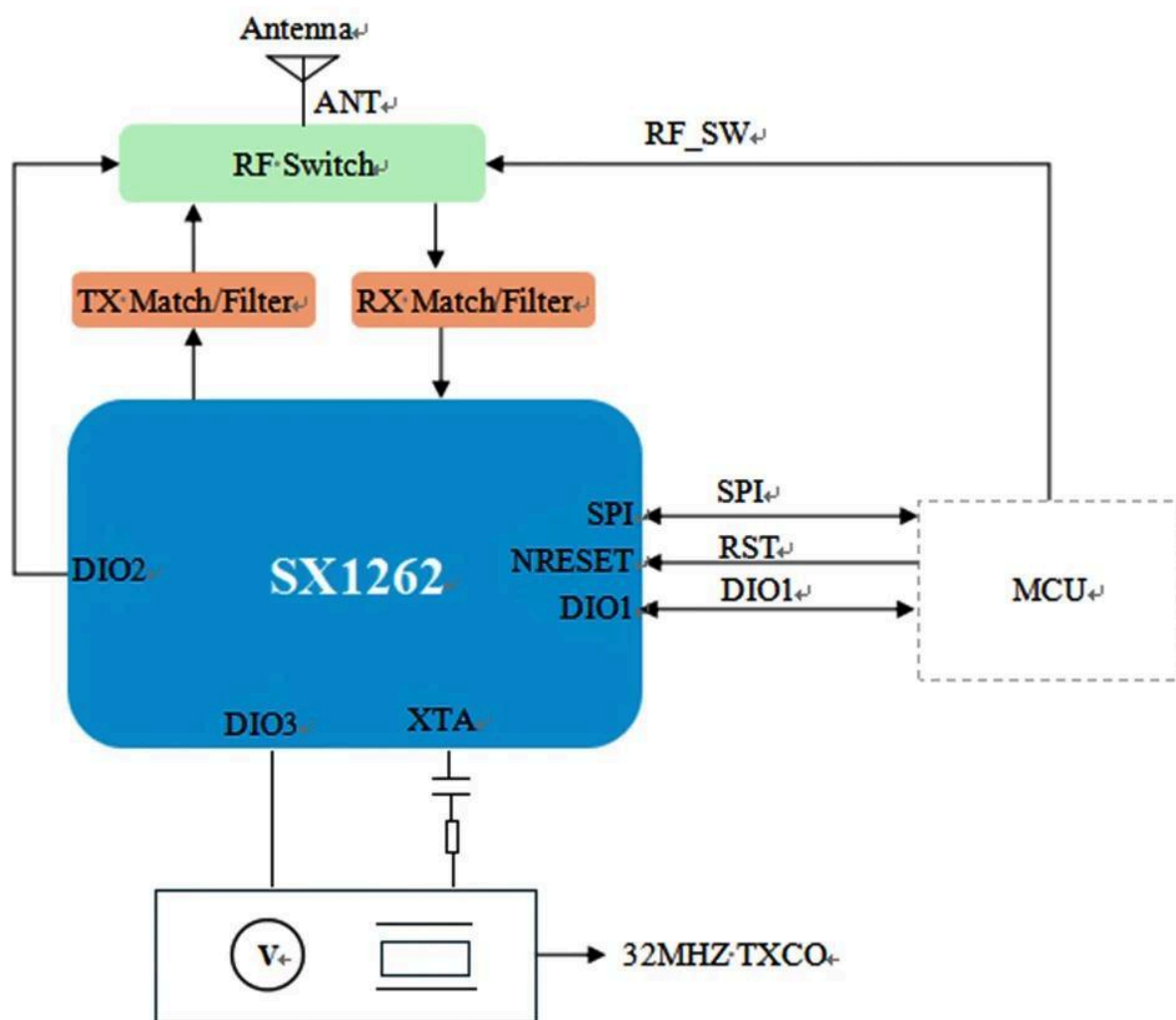
Expected Range

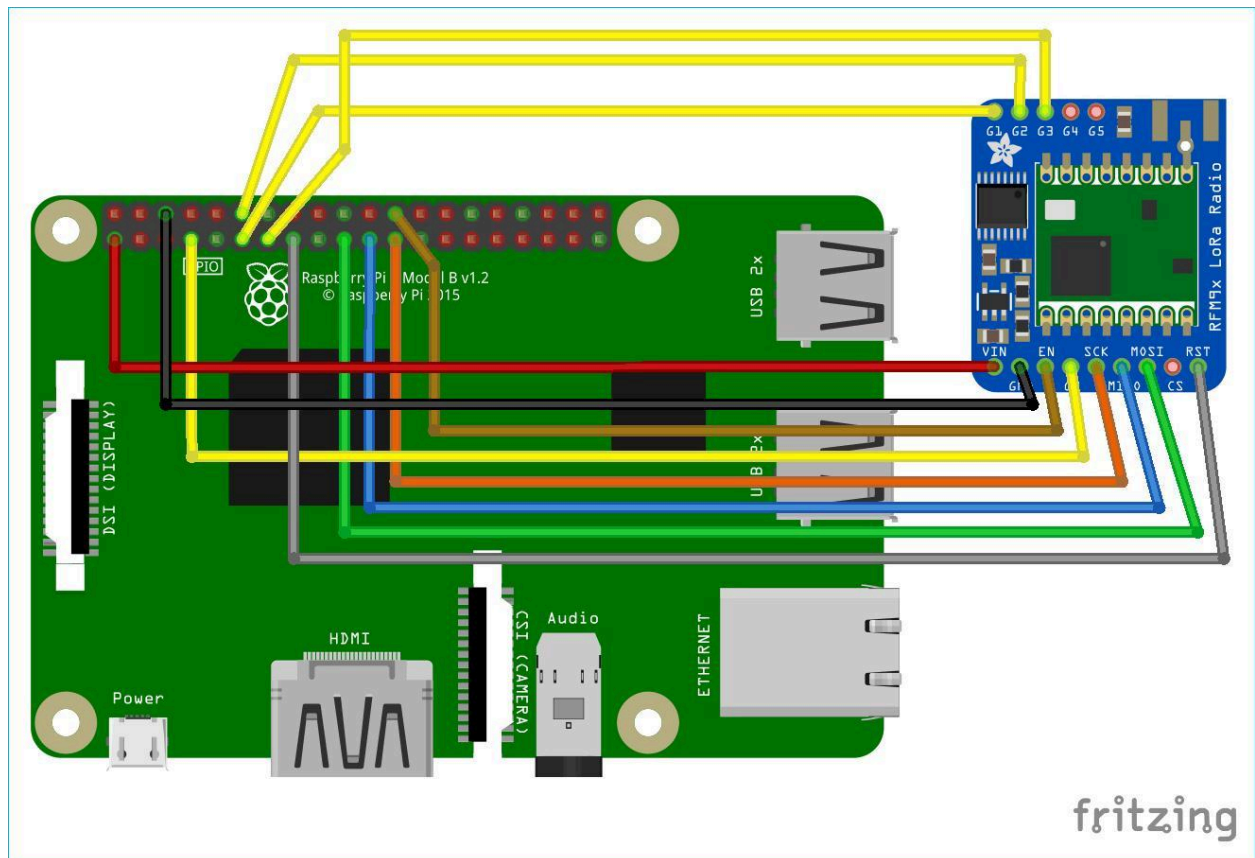
Environment	Range
City	2–5 km
Village/Open field	10+ km

([Zbotic](#))

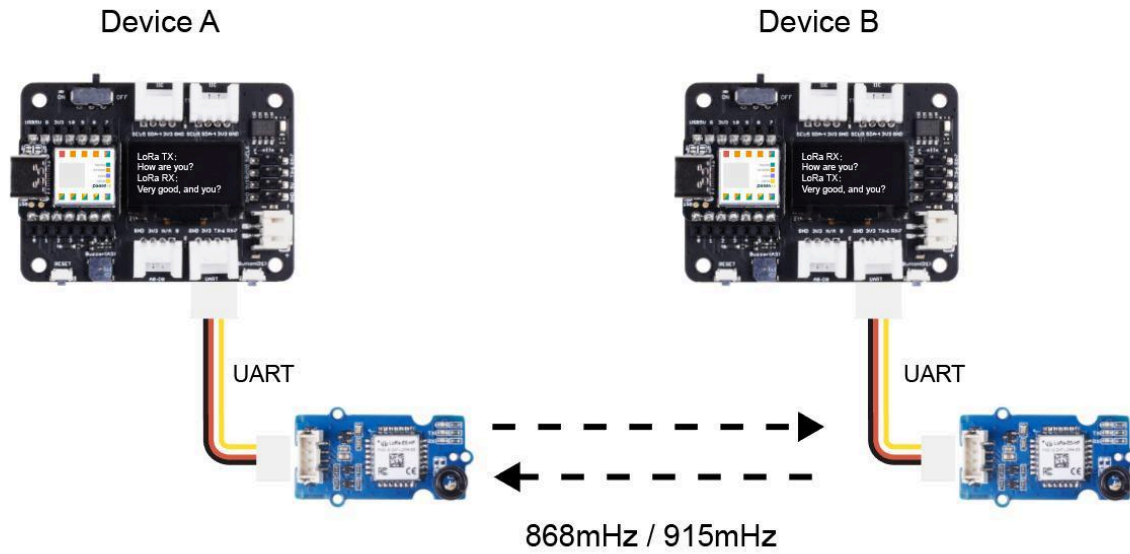
Simple Communication Diagram

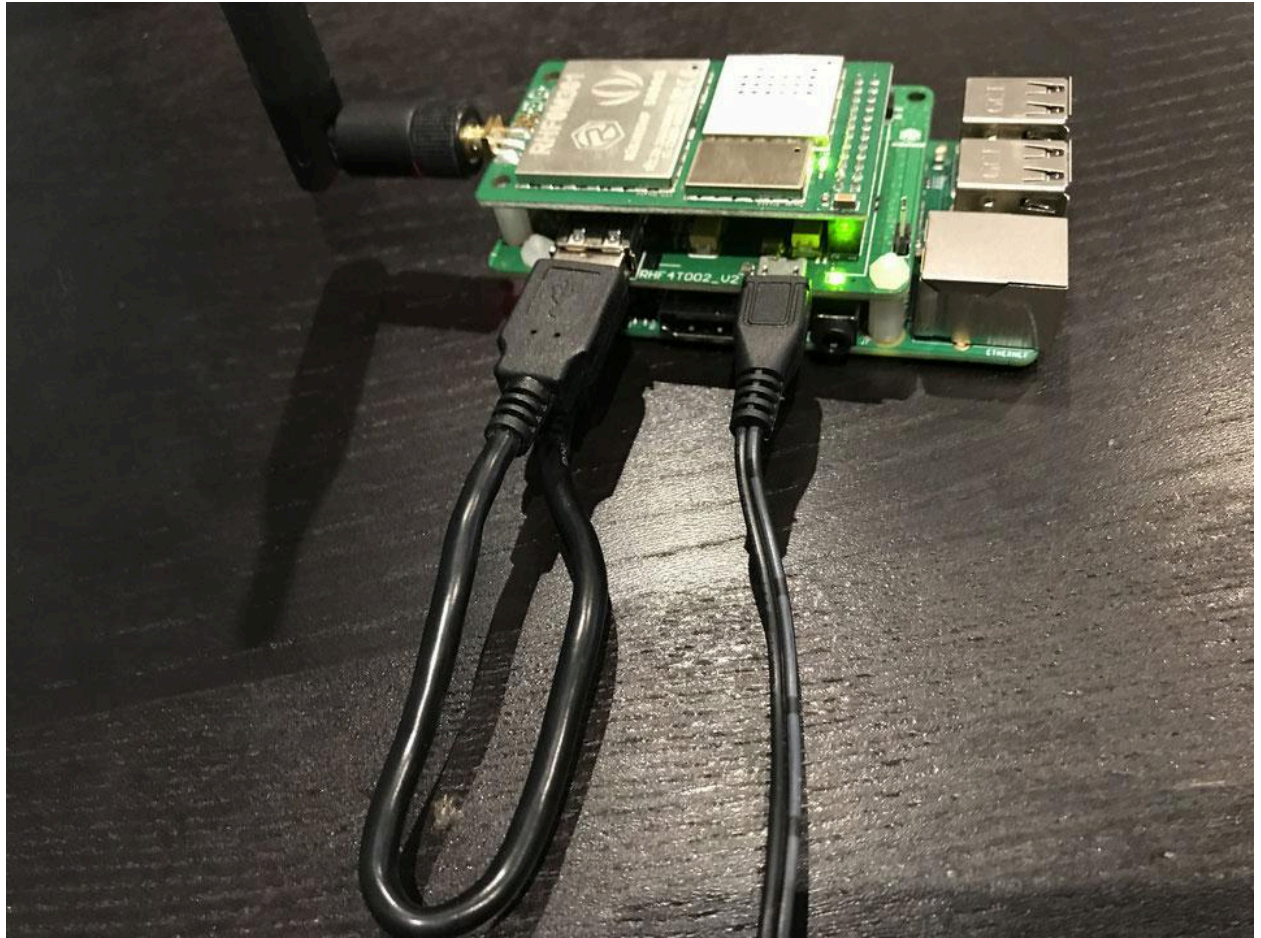






Point-to-Point Transmission with Grove - LoRa-E5







[Raspberry Pi 5] ←LoRa→ [Raspberry Pi 5]
Sender Receiver

Common Problems

No Message Received

- Check antenna connected
- Same frequency on both devices
- SPI enabled
- Correct wiring

Garbage Characters

Usually baud-rate or configuration mismatch. Reddit users reported this often with wrong jumper settings. ([Reddit](#))

Better Option

If you want:

- mobile messaging
- mesh networking
- GPS sharing

then look at:

Meshtastic

Many people use SX1262 + Raspberry Pi for Meshtastic long-range messaging. ([Reddit](#))